# Project Member Tasks

# SUBLIME

2025-26

## Mathematics Club


Centre for Innovation
IIT Madras

### Week-1 tasks

1. Familiarize yourself with GitHub, Latex and Notion.
2. Please create a new folder in the following directory:

   Folder Path

   `.\Contributors\PM tasks\`*`Your Name`*`\`

   Example

   `.\Contributors\PM tasks\Deena\`

   You're supposed to add the responses to the tasks in your folder and make a PR request.

3. Divide the topics from the second chapter of the ISLR book and present in the next meeting.

### Week-2 tasks

4. Setup VS-Code and MinGW which are essential to write and execute C++ code (MinGW is not needed if you're a Linux user).
5. Learn how to write basic functions, loading a dataset and estimate few parameters in C++.
6. Take a dataset from here and load this and evaluate basic statistical parameters.

### Week-3 tasks

7. Read the **Simple** and **Multiple** Linear regression topics from ISLR book (Chapter-3).
8. Write a sub-routine in C++ with this dataset and estimate the mentioned parameters.
9. Implement Forward stepwise selection (Chapter 6) and estimate how well the model performs.

### Week-4 tasks

10. Revise the lasso and ridge regression from the ISLR book and implement them.
11. Try to understand the D-optimally criterion in the IBOSS paper given in the Resources folder.

**Week-5 tasks**

12. Implement the algorithm based on D-optimal criterion with a dataset of your choice (Make sure that the dataset is large).
13. Ensure that your code is well-documented, as it will become part of the library.
14. Read and understand the theorems given in the properties of the parameter estimator given in the paper.